

厦门大学计算机系必修课

第二章

连续时间系统的时域分析

厦门大学计算机系
严 严

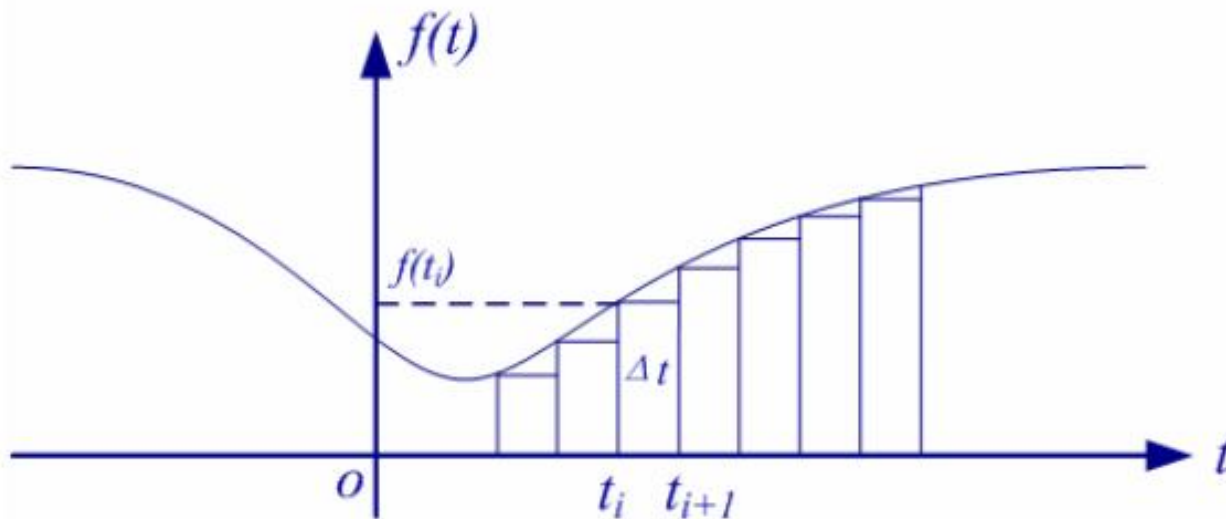
§ 2.7 卷积

回顾：任意信号可以用冲激信号的组合表示，即

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau$$

将它作用到冲激响应为 $h(t)$ 的LTI系统，则系统的响应为

$$\begin{aligned} r(t) &= H[e(t)] = H\left[\int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau\right] = \int_{-\infty}^{\infty} e(\tau) H[\delta(t - \tau)] d\tau \\ &= \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau \end{aligned}$$



$$f(t) = \lim_{\max \Delta t_i \rightarrow 0} \sum_{i=-\infty}^{\infty} f(t_i) \frac{[u(t-t_i) - u(t-t_i - \Delta t_i)]}{\Delta t_i} \Delta t_i$$

$$= \lim_{\max \Delta t_i \rightarrow 0} \sum_{i=-\infty}^{\infty} f(t_i) \delta(t-t_i) \Delta t_i$$

$$= \int_{-\infty}^{\infty} f(\tau) \delta(t-\tau) d\tau \quad (\text{此式即 } \delta(t) \text{ 的筛选特性})$$

$$\triangleq f(t) * \delta(t) \quad (1-64)$$

$\int_{-\infty}^{\infty} e(\tau)h(t-\tau)d\tau$ 就是卷积积分，用 * 标记

$$\Rightarrow r(t) = e(t) * h(t)$$

一般地，对于任意两个信号 $f_1(t)$ 和 $f_2(t)$ ，二者卷积定义为

$$f(t) = f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau)f_2(t-\tau)d\tau$$

$$\ominus \int_{-\infty}^{\infty} f_1(t-\tau)f_2(\tau)d\tau = \int_{-\infty}^{\infty} f_1(\tau)f_2(t-\tau)d\tau$$

$$\Rightarrow f_1(t) * f_2(t) = f_2(t) * f_1(t)$$

用卷积积分法求零状态响应

卷积积分：设任意信号 $f_1(t)$ 和 $f_2(t)$

$$\text{则 } f(t) = f_1(t) * f_2(t)$$

$$\xrightarrow{\text{def}} \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

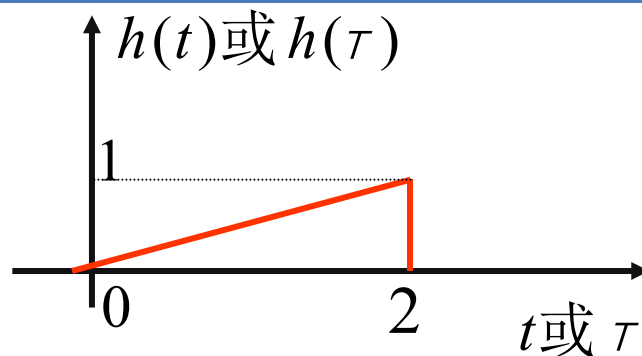
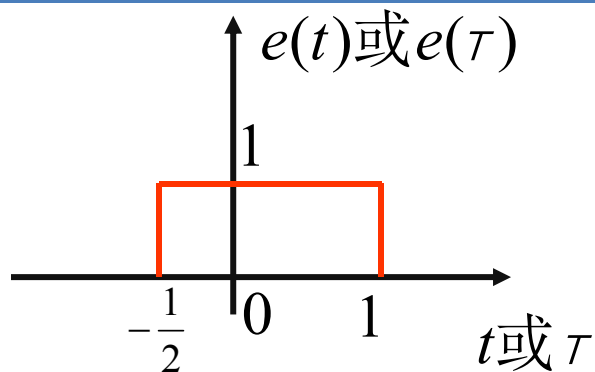
$$\text{零状态响应： } r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau = e(t) * h(t)$$

卷积积分图解法

- 卷积积分图解法：可以把卷积运算中一些抽象的关系形象化，便于理解卷积的概念及方便运算。
- 卷积积分图解法五个步骤：
 - 1、反折 2、平移 3、相乘 4、相加
- 具体地：
 - (1) 改换图形中的横坐标，由 t 改为 τ ， τ 变成函数的自变量；
 - (2) 把其中一个信号反折（反褶）。
 - (3) 把反折后的信号做位移，移位量是 t ，这样 t 是一个参变量。在 τ 坐标系中， $t>0$ 图形右移； $t<0$ 图形左移。
 - (4) 两信号重叠部分相乘 $e(\tau)h(t-\tau)$ ；
 - (5) 完成相乘后图形的积分。

举例2.10

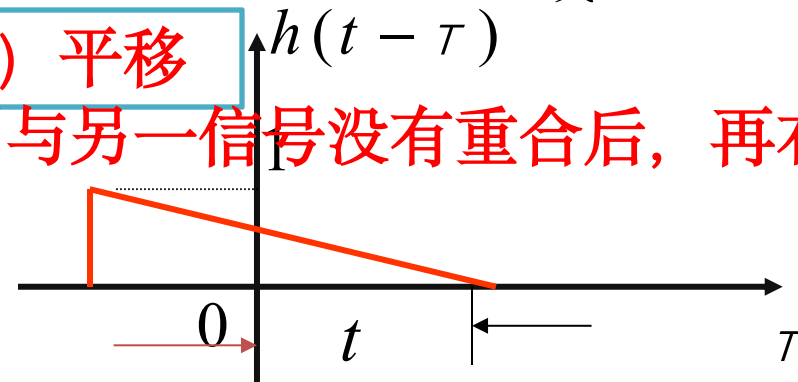
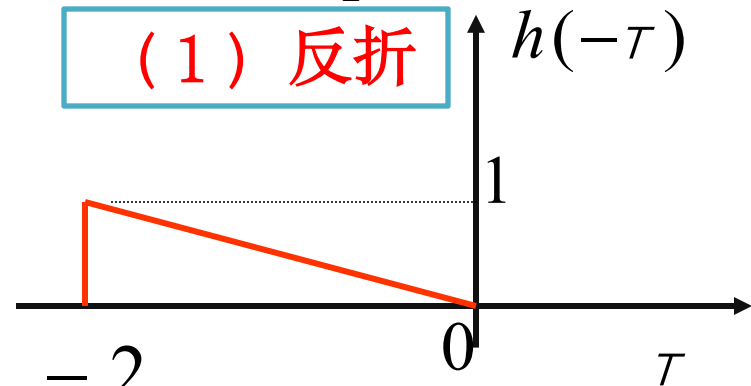
§ 2.8 卷积的性质



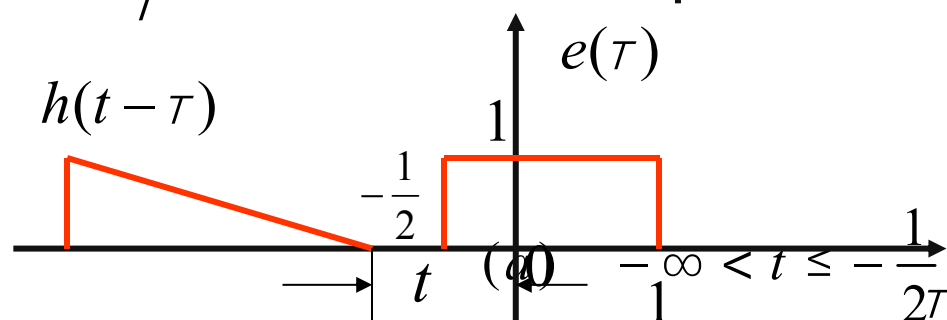
(1) 反折

(2) 平移

左移到与另一信号没有重合后，再右移。



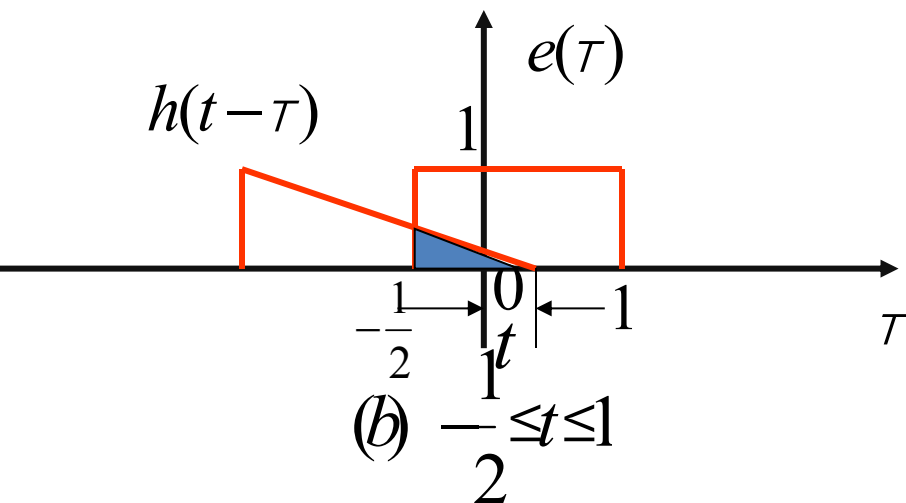
(3) 相乘



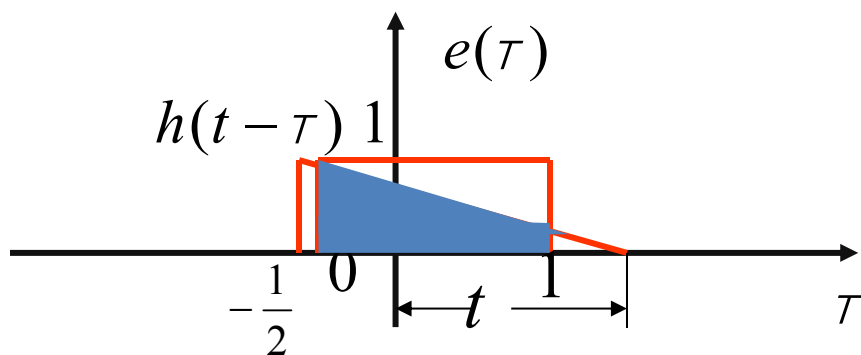
(a) $-\infty < t \leq -\frac{1}{2}$

$$e(t) * h(t) = 0$$

§ 2.8 卷积的性质



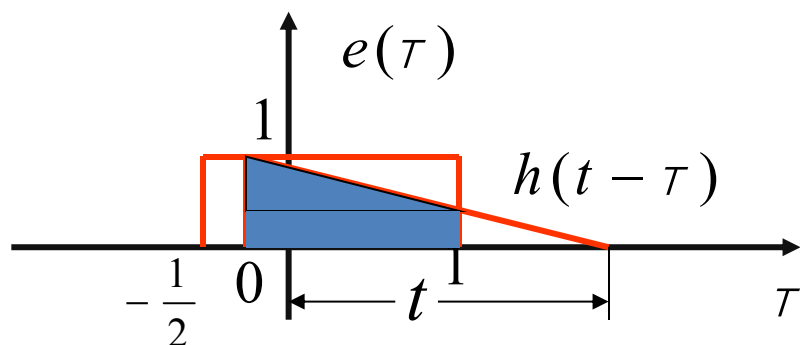
$$\begin{aligned} (b) \quad -\frac{1}{2} \leq t \leq 1 \\ e(t) * h(t) &= \int_{-\frac{1}{2}}^t 1 \times \frac{1}{2}(t - \tau) d\tau \\ &= \frac{t^2}{4} + \frac{t}{4} + \frac{1}{16} \end{aligned}$$



$$(c) \quad 1 \leq t \leq \frac{3}{2}$$

$$\begin{aligned} (c) \quad 1 \leq t \leq \frac{3}{2} \\ e(t) * h(t) &= \int_{-\frac{1}{2}}^1 1 \times \frac{1}{2}(t - \tau) d\tau \\ &= \frac{3}{4}t - \frac{3}{16} \end{aligned}$$

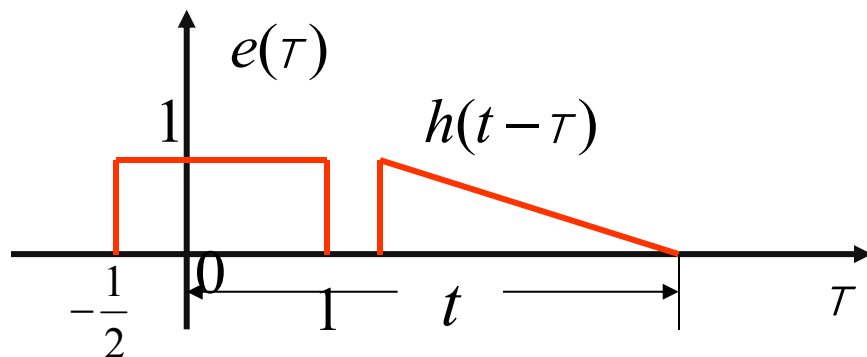
§ 2.8 卷积的性质



$$(d) \quad \frac{3}{2} \leq t \leq 3$$

$$(d) \quad \frac{3}{2} \leq t \leq 3$$

$$\begin{aligned} e(t) * h(t) &= \int_{t-2}^1 1 \times \frac{1}{2}(t-\tau) d\tau \\ &= -\frac{t^2}{4} + \frac{t}{2} + \frac{3}{4} \end{aligned}$$

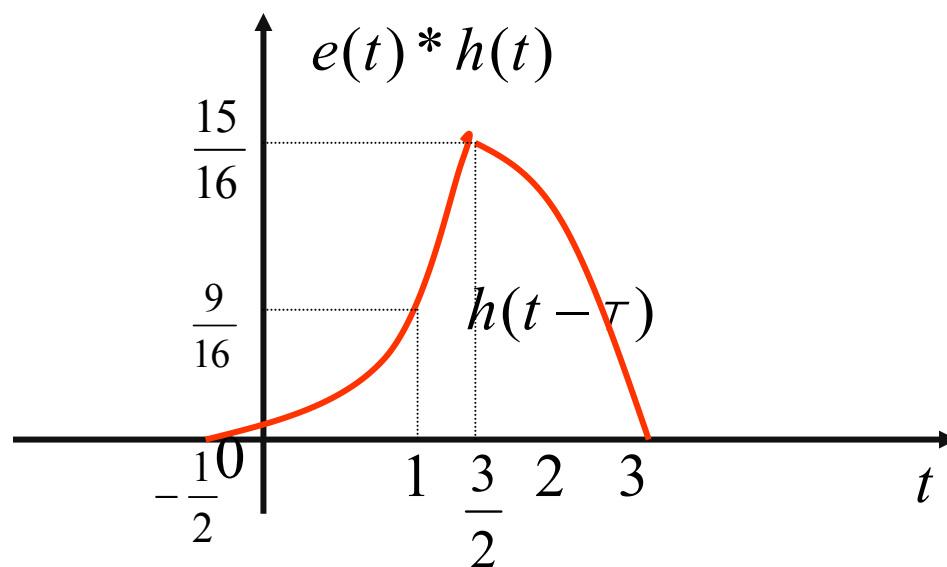


$$(e) \quad 3 < t \leq \infty$$

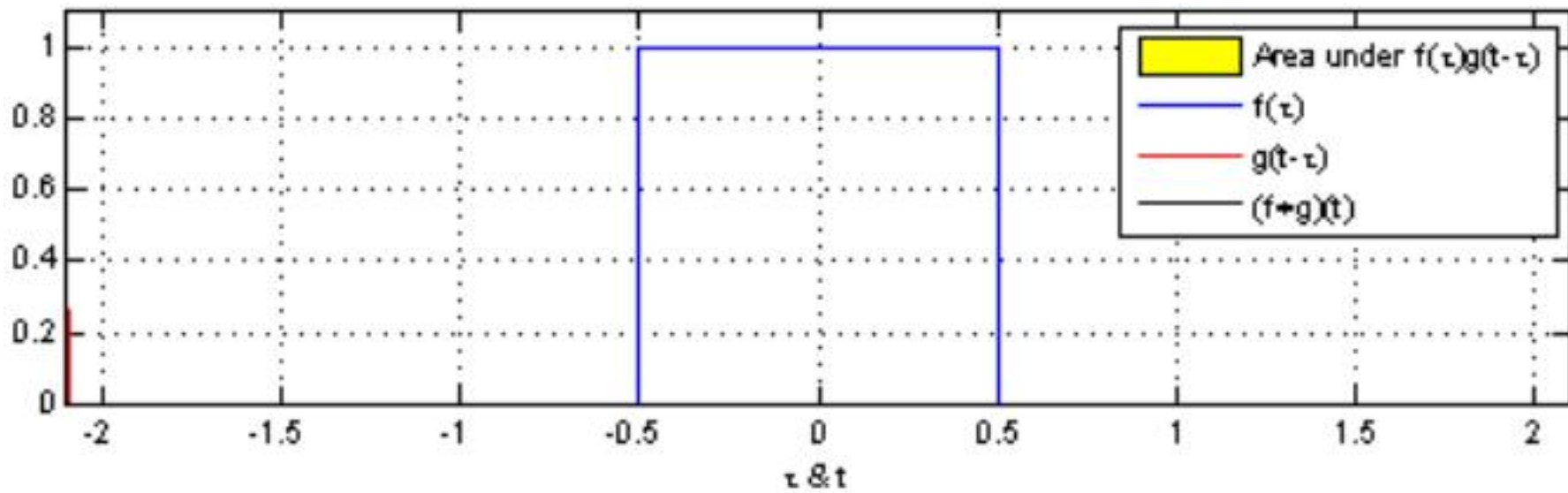
$$(e) \quad 3 < t \leq \infty$$

$$e(t) * h(t) = 0$$

(4) 相加：以上各图中的阴影面积，即为相乘积分的结果。最后，若以 t 为横坐标，将与 t 对应积分值描成曲线，就是卷积积分 $e(t)*h(t)$ 函数图像。



卷积积分结果



§ 2.8 卷积的性质

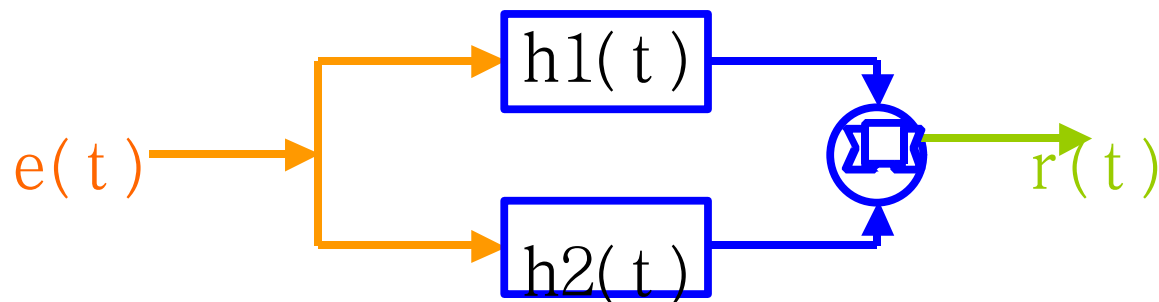
- 作为一种数学运算，卷积运算具有某些特殊性质，这些性质在信号分析中有重要作用。
- 卷积性质可以使卷积运算简化。

1.卷积代数

(1) 交换律: $f_1(t) * f_2(t) = f_2(t) * f_1(t)$

(2) 分配律: $f_1(t) * [f_2(t) + f_3(t)] =$
 $f_1(t) * f_2(t) + f_1(t) * f_3(t)$

卷积的性质



并联系统 $r(t) = e(t) * [h_1(t) + h_2(t)]$

(3) 结合律: $[f_1(t) * f_2(t)] * f_3(t) = f_1(t) * [f_2(t) * f_3(t)]$



串联系统 $r(t) = e(t) * [h_1(t) * h_2(t)]$

2.卷积的微分与积分

$$(4) \text{ 微分性: } \frac{d[f_1(t) * f_2(t)]}{dt} = f_1(t) * \frac{df_2(t)}{dt}$$

$$\text{(适于高阶微分)} \quad = \frac{df_1(t)}{dt} * f_2(t)$$

$$(5) \text{ 积分性: } \int_{-\infty}^t [f_1(\tau) * f_2(\tau)] d\tau = f_1(t) * \int_{-\infty}^t f_2(\tau) d\tau$$

$$\text{(适于多重积分)} \quad = \int_{-\infty}^t f_1(\tau) d\tau * f_2(t)$$

(6) 微积分性: 设 $f(t) = f_1(t) * f_2(t)$

$$\text{则 } f^{(i)}(t) = f_1^{(j)}(t) * f_2^{(i-j)}(t)$$

$$\text{如 } f_1(t) * f_2(t) = f_1'(t) * f_2^{(-1)}(t)$$

$$f^{(2)}(t) = f_1'''(t) * f_2^{(-1)}(t)$$

$$f^{(-2)}(t) = f_1''(t) * f_2^{(-4)}(t)$$

与冲激函数或阶跃函数的卷积

(7) 冲激性: $f(t) * \delta(t) = f(t)$

$$f(t) * \delta(t - t_0) = f(t - t_0)$$

$$f(t) * \delta^{(k)}(t) = f^{(k)}(t)$$

$$f(t) * \delta^{(k)}(t - t_0) = f^{(k)}(t - t_0)$$

($k > 0$ 时为导数阶次,

$k < 0$ 时为积分次数)

(8) 阶跃性: $f(t) * u(t) = f^{(-1)}(t) = \int_{-\infty}^t f(\tau) d\tau$

几种特例

$$\delta(t) * \delta(t) = \delta(t)$$

$$\delta(t) * \delta(t - a) = \delta(t - a)$$

$$\delta(t - a) * \delta(t - b) = \delta(t - a - b)$$

卷积的性质可以用来**简化卷积运算**

举例2.11:

$$\text{已知 } f_1(t) = (1+t)[u(t) - u(t-1)], \\ f_2(t) = u(t-1), \text{ 求 } f_1(t) * f_2(t)$$

$$\begin{aligned} \text{解: } f_1(t) * f_2(t) &= \int_{-\infty}^t f_1(\tau) d\tau * \frac{df_2(t)}{dt} \quad (\text{微积分性}) \\ &= \int_{-\infty}^t (1+\tau)[u(\tau) - u(\tau-1)] d\tau * \delta(t-1) \\ &= \left[\int_0^t (1+\tau) d\tau - \int_1^t (1+\tau) d\tau \right] * \delta(t-1) \end{aligned}$$

$$\begin{aligned}
 f_1(t) * f_2(t) &= \left[\int_0^t (1 + \tau) d\tau - \int_1^t (1 + \tau) d\tau \right] * \delta(t - 1) \\
 &= \left[\tau + \frac{\tau^2}{2} \Big|_0^t - \tau + \frac{\tau^2}{2} \Big|_1^t \right] * \delta(t - 1) \\
 &= \left[\left(t + \frac{t^2}{2} \right) u(t) - \left(t - 1 + \frac{t^2 - 1}{2} \right) u(t - 1) \right] * \delta(t - 1)
 \end{aligned}$$

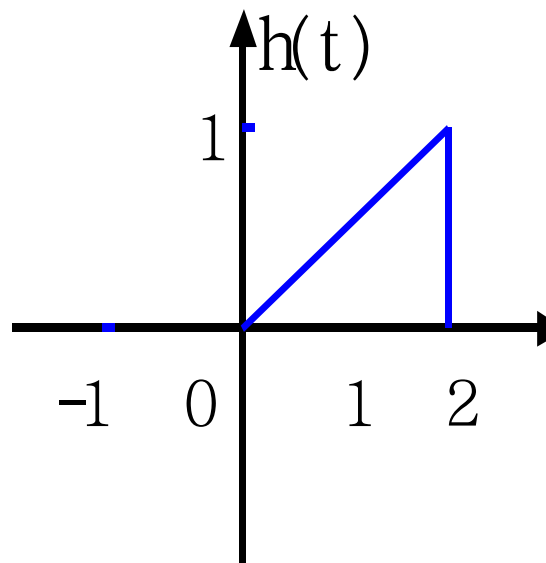
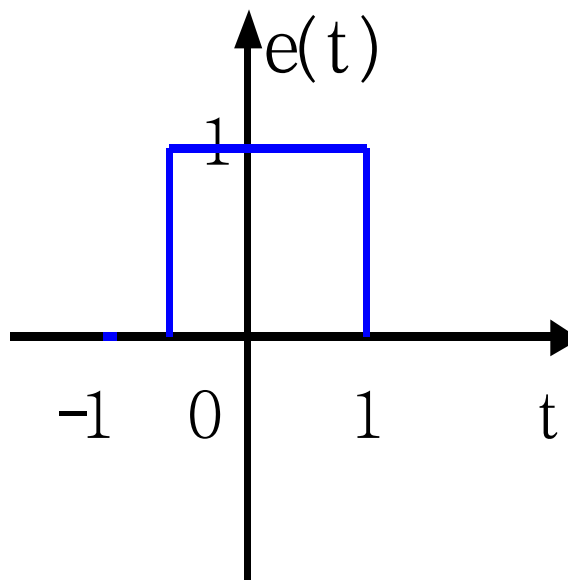
$$f_1(t) * f_2(t) = \left[t - 1 + \frac{(t-1)^2}{2} \right] u(t-1) - \left[t - 2 + \frac{(t-1)^2 - 1}{2} \right] u(t-2)$$

$$= \frac{t^2 - 1}{2} u(t-1) - \frac{t^2 - 4}{2} u(t-2)$$

(卷积的冲激性)

举例2.12:

如图所示系统的 $e(t)$ 、 $h(t)$ ，求其零状态响应



$$e(t) = u\left(t + \frac{1}{2}\right) - u(t - 1) \quad h(t) = \frac{1}{2}t[u(t) - u(t - 2)]$$

• 解: $e(t) * h(t) = \frac{de(t)}{dt} * \int_{-\infty}^t h(\tau) d\tau$ (微积分性)

$$\therefore \frac{de(t)}{dt} = \delta\left(t + \frac{1}{2}\right) - \delta(t - 1)$$

$$\begin{aligned}\int_{-\infty}^t h(\tau) d\tau &= \frac{1}{2} \int_{-\infty}^t \tau [u(\tau) - u(\tau - 2)] d\tau \\ &= \frac{1}{2} \left(\int_0^t \tau d\tau - \int_2^t \tau d\tau \right) \\ &= \frac{1}{4} t^2 [u(t) - u(t - 2)] + u(t - 2)\end{aligned}$$

$$\therefore \mathbf{e}(t) * \mathbf{h}(t) = \left[\delta\left(t + \frac{1}{2}\right) - \delta(t - 1) \right] * \left\{ \frac{1}{4} t^2 [\mathbf{u}(t) - \mathbf{u}(t - 2)] + \mathbf{u}(t - 2) \right\}$$

(卷积的分配律)

$$\begin{aligned} &= \delta\left(t + \frac{1}{2}\right) * \left\{ \frac{1}{4} t^2 [\mathbf{u}(t) - \mathbf{u}(t - 2)] \right\} + \delta\left(t + \frac{1}{2}\right) * \mathbf{u}(t - 2) \\ &\quad - \delta(t - 1) * \left\{ \frac{1}{4} t^2 [\mathbf{u}(t) - \mathbf{u}(t - 2)] \right\} - \delta(t - 1) * \mathbf{u}(t - 2) \end{aligned}$$

$$\begin{aligned} e(t) * h(t) = & \left\{ \frac{1}{4} t^2 [u(t) - u(t-2)] \right\} * \delta\left(t + \frac{1}{2}\right) + u(t-2) * \delta\left(t + \frac{1}{2}\right) \\ & - \left\{ \frac{1}{4} t^2 [u(t) - u(t-2)] \right\} * \delta(t-1) - u(t-2) * \delta(t-1) \end{aligned}$$

(卷积的交换律)

(卷积的冲激性)

$$e(t) * h(t) = \frac{1}{4} \left(t + \frac{1}{2} \right)^2 \left[u \left(t + \frac{1}{2} \right) - u \left(t - \frac{3}{2} \right) \right] + u \left(t - \frac{3}{2} \right)$$

$$- \frac{1}{4} (t - 1)^2 [u(t - 1) - u(t - 3)] - u(t - 3)$$

$$\therefore r_{zs}(t) = \begin{cases} \frac{1}{4} \left(t + \frac{1}{2} \right)^2 & -\frac{1}{2} \leq t < 1 \\ \frac{3}{4} \left(t - \frac{1}{4} \right) & 1 \leq t < \frac{3}{2} \\ 1 - \frac{1}{4} (t - 1)^2 & \frac{3}{2} \leq t < 3 \end{cases}$$

课后自学: § 2.9, § 2.10

[注意]①通过分配函数的概念加深对 $\delta(t)$ 性质的理解, 记住若干有用的性质

如 $\delta(at) = \frac{1}{|a|} \delta(t)$ 等

第一、二章小结

基本概念(信号、系统)

奇异信号 (冲激、阶跃) 、线性时不变系统

连续时间系统时域分析方法 (微分方程求解、卷积)

课后作业: 2-13, 2-15, 2-18